PHYS 225

Week 13, Lecture 11:
Relativistic collisions, decays, and reactions

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Week 13, Lecture 11

Relativistic collisions, decays, and reactions

Objectives:

- Use natural units to express masses and momenta in terms of energy
- Use conservation of energy and momentum to determine the kinematics of relativistic reactions
- Use invariance of the relativistic dot product and the center-of-momentum frame to enormously simplify calculations!

One Minute Paper: write down the energy-momentum 4-vector for a particle of mass m in its own rest frame, in natural units where c = 1.

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Natural units, again

The standard unit for energy in nuclear and particle physics is the "electron-volt", eV.

Definition: if an electron at rest traverses a potential difference of 1 V, it will gain kinetic energy of 1 eV.

$$\boldsymbol{E_k = qU \;\; \rightarrow \; 1\ \mathrm{eV} = 1.6\ \ 10^{-19}\ \mathrm{C} \times 1\ \mathrm{V} = 1.6\ 10^{-19}\mathrm{J}}$$

Common prefixes:
keV = 10^3 eV
MeV = 10^6 eV
GeV = 10^9 eV

$$E_0 = mc^2 \Longleftrightarrow E_0 = m$$

$$m_e = 511\ \mathrm{keV}$$

$$m_p = 938\ \mathrm{MeV}$$

Average **daily electricity consumption** in C-U **~10^{27}** eV per household.
A typical **car moving at 70 mph** carries about **~10^{24}** eV of kinetic energy.

In this lecture, all energies, momenta, and masses have eV units.
To get back to SI, momenta are eV/c and masses are eV/c^2

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Relativistic reactions

Special relativity + quantum mechanics:
particles can decay into other particles!

$$\pi^0 \rightarrow \gamma + \gamma$$

pion decay

$$n \rightarrow p + e^- + \bar{\nu}_e$$

beta decay

$${}^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$$

alpha decay

Or, make new particles by smashing other particles together!

$$e^+ + e^- \rightarrow \gamma + \gamma$$

matter-antimatter annihilation

$$p + \bar{p} \rightarrow H + \ldots$$

Higgs boson discovery

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Relativistic kinematics — tips

1. Velocities and ugly square roots are NOT your friends. There is almost ALWAYS a better way! Energy and momentum are the right variables.

2. Use invariance of the relativistic dot product: compute the same quantity in two different frames, and set the expressions equal (because the result must be the same in all frames)

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Kinematics of particle decay

A charged pion of mass M decays most of the time into a muon (mass m) and a neutrino (which we approximate as massless).
What is the energy of the outgoing neutrino in the rest frame of the pion?

$$\pi^+ \rightarrow \mu^+ + \nu$$

First step is **always** to assign energy-momentum 4-vectors to each particle:

$$p_1^\mu = p_2^\mu + p_3^\mu$$

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Kinematics of particle decay

Next, let's assign the components of the 4-vectors in some frame. In this case, the obvious frame is the pion rest frame.

$$p_1 = (M, \vec{0})$$

$$p_2 = (E_2, \vec{p}_2)$$

$$p_3 = (E_3, \vec{p}_3)$$

What do we want? E_3

What do we **not** care about? The directions of p_2 and p_3, the muon energy E_2, or the velocity of any particle

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Kinematics of particle decay

THE trick to use whenever you can: rearrange the 4-momentum conservation equation and take convenient (relativistic) dot products

$$p_1 = (M, \vec{0})$$

$$p_2 = (E_2, \vec{p}_2)$$

$$p_3 = (E_3, \vec{p}_3)$$

$$p_1 = p_2 + p_3$$

$$p_1 - p_3 = p_2$$

$$(p_1 - p_3) \cdot (p_1 - p_3) = p_2 \cdot p_2$$

$$p_1 \cdot p_1 + p_3 \cdot p_3 - 2p_1 \cdot p_3 = p_2 \cdot p_2$$

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Kinematics of particle decay

This is useful because the dot product of any massive 4-vector with itself is m^2 (this is true in the rest frame, and is also guaranteed by $E^2 - p^2 = m^2$)

$$p_1 = (M, \vec{0}) \qquad p_1 \cdot p_1 = M^2$$

$$p_2 = (E_2, \vec{p}_2) \qquad p_2 \cdot p_2 = m^2$$

$$p_3 = (E_3, \vec{p}_3) \qquad p_3 \cdot p_3 = 0$$ (neutrino is massless)

$$p_1 \cdot p_1 + p_3 \cdot p_3 - 2p_1 \cdot p_3 = p_2 \cdot p_2$$

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Kinematics of particle decay

Why did we pick p_3 to move to the other side? The one nontrivial dot product we have is easy in the rest frame of p_1:

$$p_1 = (M, \vec{0})$$

$$p_2 = (E_2, \vec{p}_2) \qquad p_1 \cdot p_3 = ME_3 - \vec{0} \cdot \vec{p}_3 = ME_3$$

$$p_3 = (E_3, \vec{p}_3)$$

$$p_1 \cdot p_1 + p_3 \cdot p_3 - 2p_1 \cdot p_3 = p_2 \cdot p_2$$

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Kinematics of particle decay

Now our work is done:

$$p_1 \cdot p_1 + p_3 \cdot p_3 - 2p_1 \cdot p_3 = p_2 \cdot p_2$$

$$M^2 - 2ME_3 = m^2$$

$$\implies E_3 = \frac{M^2 - m^2}{2M}$$

Sanity checks:

- answer has the right dimensions of energy
- if m = 0, both decay products are massless, and this is just like the example from Discussion 8: $E_3 = M/2$
- if m > M, we get a nonsensical negative answer: decaying into a heavier particle would require exceeding the original rest energy!

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Kinematics of particle production

A high-energy proton can collide with another proton at rest to produce antiprotons:

$$p + p \rightarrow p + p + p + \bar{p}$$

The antiproton has the same mass m_p as the proton.

What is the minimum amount of energy the incoming proton must have for this reaction to occur?

Key strategy: use both the **laboratory frame** and the **center-of-momentum (CM)** frame, where the total momentum is zero

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Kinematics of particle production

$$p + p \to p + p + p + \bar{p}$$

$$p_1^\mu + p_2^\mu = k_1^\mu + k_2^\mu + k_3^\mu + k_4^\mu$$

In laboratory frame:

$$p_1 = (E_{\rm lab}, \vec{p}_{\rm lab}) \qquad p_1 \cdot p_1 = p_2 \cdot p_2 = m_p^2$$

$$p_2 = (m_p, \vec{0}) \qquad p_1 \cdot p_2 = E_{\rm lab} m_p - \vec{p}_{\rm lab} \cdot \vec{0} = E_{\rm lab} m_p$$

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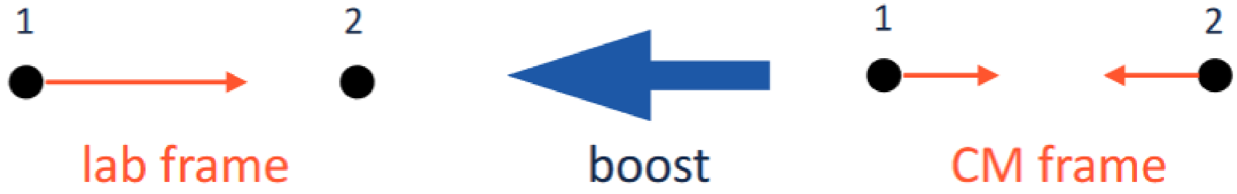
Kinematics of particle production

$$p + p \to p + p + p + \bar{p}$$

$$p_1^\mu + p_2^\mu = k_1^\mu + k_2^\mu + k_3^\mu + k_4^\mu$$

Now consider boosting to the CM frame, where the initial protons have equal and opposite momenta:



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Kinematics of particle production

We could find the boost velocity which transforms the lab frame to the CM frame. But for this problem, we don't have to!

CM frame,
initial state

$$\vec{p}_1 + \vec{p}_2 = 0$$

CM frame,
final state

$$\vec{k}_1 + \vec{k}_2 + \vec{k}_3 + \vec{k}_4 = 0$$

conservation of momentum!

The total momentum of the final state is always zero, but the minimum energy is when all of the momenta are individually zero

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Kinematics of particle production

Now apply 4-momentum conservation and invariance of the dot product:

$$p_1 + p_2 = k_1 + k_2 + k_3 + k_4$$

$$= (4m_p, \vec{0}) \text{ in CM frame}$$

$$(p_1 + p_2)^2 = (k_1 + k_2 + k_3 + k_4)^2$$

evaluate in lab frame:

$$p_1 \cdot p_1 + p_2 \cdot p_2 + 2p_1 \cdot p_2 = 2m_p^2 + 2E_{\text{lab}} m_p$$

evaluate in CM frame:

$$16m_p^2 - \vec{0} \cdot \vec{0} = 16m_p^2$$

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Kinematics of particle production

As with the decay example, we now have a simple linear equation to solve for E_{lab}:

$$2m_p^2 + 2E_{\rm lab}m_p = 16m_p^2$$
$$\implies E_{\rm lab} = 7m_p$$

Despite only having to create $2m_p$ worth of rest energy, we had to supply an additional $3m_p$ (= $7m_p$ - $4m_p$) of kinetic energy.

For particle production near threshold, a fixed target setup is kinematically more expensive than a head-on collider.

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Tricks for velocity

If you absolutely must know the velocity or boost factor: $P^\mu = (E, \vec{p}) = (\gamma m, \gamma m \vec{v})$

$$\implies \gamma = \frac{E}{m} = \frac{E}{\sqrt{P \cdot P}}$$

$$\implies \vec{v} = \frac{\vec{p}}{E}$$

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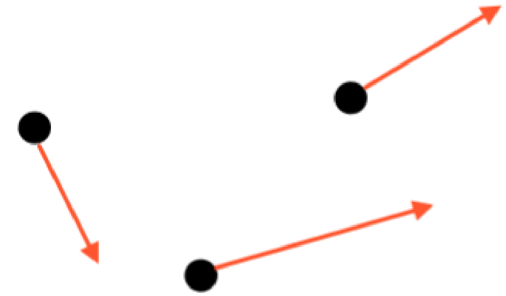
Tricks for velocity

This is useful to find the velocity of the CM frame:



$$\begin{aligned} P^{\mu}_{\text{tot.}} &= p^{\mu}_1 + p^{\mu}_2 + \dots \\ &= (E_{\text{tot.}}, \vec{p}_{\text{tot.}}) \end{aligned}$$

$$\vec{v}_{\text{CM}} = \frac{\vec{p}_{\text{tot.}}}{E_{\text{tot.}}}$$

Very easy!
Works because
sum of 4-vectors
is also a 4-vector

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Relativistic collisions, decays, and reactions

Summary

- In natural units, mass, momentum, and energy all have eV units
- Total 4-momentum in a collision or decay is conserved
- If particle 1 is at rest and you want the energy of particle 2, use the invariant dot product of 4-momenta 1 and 2
- The center-of-momentum frame is a good frame to compute dot products in
- Avoid velocity! No horrible square roots!

One Minute Paper: in our first example of pion decay, how would you rearrange p1 = p2 + p3 to find the energy of p2 (the muon)?